HDOC Day-47

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Algorithm

1. Start.
2. Read the size `n` of the array.
3. Validate that `n > 0`.
4. Declare a variable-length integer array `arr[n]`.
5. Read `n` elements in ascending sorted order.
6. Read the integer `target`.
7. Pass the address of `arr[]`, its size, and `target` to `ceil_index()`.
8. In `ceil_index()`:
   - Set `low = 0`, `high = n - 1`, and `result = -1`.
   - Use binary search while `low <= high`.
   - Calculate `mid = low + (high - low) / 2`.
   - If `*(arr + mid) >= target`:
     - Store `mid` in `result`.
     - Continue searching toward the left by setting `high = mid - 1`.
   - Otherwise, set `low = mid + 1`.
9. After the search:
   - If `result == -1`, display `-1`.
   - Otherwise, display the ceil element and its index.
10. Stop.

Test-Case Table

Test	Array	Target	Expected Output
1	5 7 7 8 8 10	6	Ceil of 6: 7 at index 1
2	5 7 7 8 9 10	8	Ceil of 8: 8 at index 3
3	5 7 7 8 8 10	11	-1
4	5 7 7 8 8 10	5	Ceil of 5: 5 at index 0
5	5 7 7 8 8 10	7	Ceil of 7: 7 at index 1
6	5 7 7 8 8 10	9	Ceil of 9: 10 at index 5

7	5 7 7 8 8 10	1	Ceil of 1: 5 at index 0

C Program

```
#include <stdio.h>

int ceil_index(int *arr, int n, int target)

{

    int low = 0;

    int high = n - 1;

    int mid;

    int result = -1;

    while (low <= high)

    {

        mid = low + (high - low) / 2;

        if (*(arr + mid) >= target)

        {

            result = mid;

         // Search towards the left for the first

            // occurrence of the smallest valid element

            high = mid - 1;

        }

        else

        {
```

```
            low = mid + 1;
        }
    }
    if (result == -1)
    {
        printf("-1\n");
    }
    else
    {
        printf("Ceil of %d: %d at index %d\n",
                target, *(arr + result), result);
    }
    return result;
}
int main()
{
    int n, target;
    printf("Enter the size of the array: ");
    scanf("%d", &n);
    while (n <= 0)
    {
        printf("Invalid size. Enter a positive size: ");
```

```
        scanf("%d", &n);
    }
    int arr[n];
    printf("Enter %d elements in ascending order:\n", n);
    for (int i = 0; i < n; i++)
    {
        scanf("%d", arr + i);
    }
    printf("Enter target: “);
    scanf("%d", &target);
    ceil_index(arr, n, target);
   return 0;
}
```

EXECUTION

```
[aaryanrajput@aaryans-macbook ~ % nano day47.c
[aaryanrajput@aaryans-macbook ~ % clang day47.c -o day47.c
[aaryanrajput@aaryans-macbook ~ % ./day47.c
 Enter the size of the array: 6
 Enter 6 elements in ascending order:
 5 7 7 8 9 11
 Enter target: 6
 Ceil of 6: 7 at index 1
[aaryanrajput@aaryans-macbook ~ % ./day47.c
 Enter the size of the array: 6
 Enter 6 elements in ascending order:
 5 6 7 8 11 21
 Enter target: 8
 Ceil of 8: 8 at index 3
[aaryanrajput@aaryans-macbook ~ % ./day47.c
 Enter the size of the array: 5
 Enter 5 elements in ascending order:
 5 7 7 8 11
 Enter target: -1
 Ceil of -1: 5 at index 0
 aaryanrajput@aaryans-macbook ~ %
```